

Chapter 2. PERMUTATIONS

2.2 Cycles



Today we will learn how to **factor a permutation into a product of disjoint cycles**

How to factor a permutation into a product of disjoint cycles

Consider the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & \underline{5} & 6 & 7 & 8 & 9 \\ 6 & 4 & 7 & 2 & \underline{5} & 1 & 8 & 9 & 3 \end{pmatrix} \in S_9$$

$$\sigma = (1 \ 6)(2 \ 4)(3 \ 7 \ 8 \ 9)$$

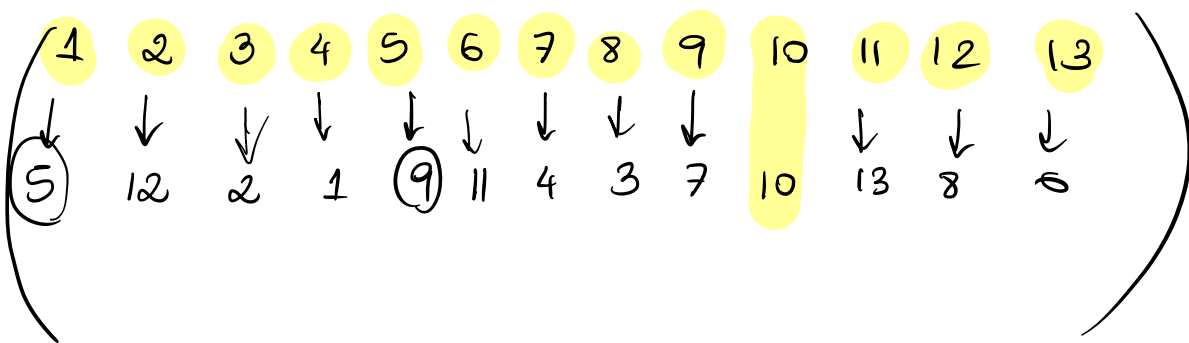
Look at $\sigma(1) = 6$, and then we find $\sigma(6) = 1$, so we obtain with cycle. Now we look at the first element in $\{1, 2, 3, 4, 5, 6\}$ that is different from 1 and 6.

$$\begin{pmatrix} 1 & \textcircled{2} & \textcircled{3} & \textcircled{4} & 5 & \textcircled{6} & \textcircled{7} & \textcircled{8} & \textcircled{9} \\ \downarrow & \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow & \downarrow & \downarrow \\ \textcircled{6} & \underline{4} & 7 & \underline{2} & 5 & 1 & \textcircled{8} & 9 & 3 \end{pmatrix}$$

Example 2.15. Factor

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \\ 5 & 12 & 2 & 1 & 9 & 11 & 4 & 3 & 7 & 10 & 13 & 8 & 6 \end{pmatrix} \in S_{13}$$

as a product of (pairwise) disjoint cycles.



$$\sigma = (1 \ 5 \ 9 \ 7 \ 4)(2 \ 12 \ 8 \ 3)(6 \ 11 \ 13)$$

Remark 2.16.

σ and τ are disjoint cycles $\Leftrightarrow \{k_1, k_2, \dots, k_r\} \cap \{\ell_1, \ell_2, \dots, \ell_s\} = \emptyset$,

where $\sigma = (k_1 \ k_2 \ \dots \ k_r)$ and $\tau = (\ell_1 \ \ell_2 \ \dots \ \ell_s)$.

